MIL-B-62542(AT)
20 January 1988
SUPERSEDING
(see 6.5)

MILITARY SPECIFICATION

BALL SCREW LINEAR ACTUATOR

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a Ball Screw Linear Actuator (BSLA) for deployment of stabilizing outriggers for the M860A1 semitrailers. The BSLA is capable of 52,000 pounds (lbs) load over 37 inches of retraction and extension.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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SPECIFICATIONS
FEDERAL

P-C-437	- Cleaning Compound, High Pressure (Steam) Cleaner.
QQ-C-320	- Chromium Plating (Electrodeposited).
QQ-P-35	- Passivation Treatments for Corrosion-Resisting Steel.
TT-C-490	- Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings.

MILITARY

MIL-H-5606	- Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance.
MIL-F-13927	- Fungus Resistance Test; Automotive Components.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw, NATO Code Number G-354 Metric.
MIL-P-26915	- Primer Coating, Zinc Dust Pigmented, for Steel Surfaces.
MIL-A-46106	- Adhesive-Sealants, Silicone, RTV, General-Purpose.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-P-52192	- Primer, Coating, Epoxy.
MIL-P-53030	- Primer, Coating, Epoxy, Water Reducible, Lead and Chromate Free.
MIL-C-53039	- Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.

STANDARDS
MILITARY

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-193	- Painting Procedures and Marking for Vehicles, Construction Equipment and Material Handling Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems Equipment and Facilities.
MIL-STD-45662	- Calibration Systems Requirements.

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2.1.2 Government drawing. The following Government drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

DRAWING
ARMY

11686299 - Ball Screw Linear Actuator (BSLA).

(Copies of specifications, standards and drawing required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish BSLAs which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information, shall be representative of the units to be furnished to the Government. All subsequent BSLAs delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Durability profile. The durability of the BSLA shall be determined when operating the BSLA for a total of 500 operational durability cycles per unit under various environmental conditions (see 3.8).

3.2.1 Reliability and maintainability conformance. The reliability and maintainability of the BSLA shall be determined based on an evaluation of the reliability failures that occurred and the maintainability tasks performed under the above durability profile (see 3.8.8 and 3.8.9).

3.3 Materials. Materials shall be as specified herein, on applicable standards or drawings, or in applicable specifications. Materials susceptible to fungus shall meet the requirements of MIL-F-13927 (see 4.7.1).

3.3.1 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.3.1).

3.4 Design and construction.

3.4.1 Configuration. The BSLA shall conform to Drawing 11686299 (see 4.7.1 and 4.7.2).

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3.4.2 Qualified products. Parts or components that have been qualified for listing on Qualified Product Lists (QPLs) or approved for listing on QPLs shall be used in the BSLA when appropriate (see 4.7.1).

3.4.3 BSLA lubrication. The BSLA shall be lubricated with grease conforming to MIL-G-23827 (see 4.7.1 and 4.7.2).

3.4.4 Gear box lubrication. The gear box shall be filled with petroleum-based hydraulic fluid conforming to MIL-H-5606 (see 4.7.1 and 4.7.2).

3.4.5 Electromagnetic interference. The BSLA shall meet electromagnetic compatibility requirements of MIL-STD-461 for tactical vehicles, class IIC, RE02 and CE03 (see 4.7.3.1).

3.4.6 Ball and thread. The ball screws shall have a thread diameter of 2.500 inches and a lead of 0.500 inch. The balls shall have a nominal diameter of 0.375 inch. The maximum allowable backlash shall be 0.020 inch (see 4.7.3.2).

3.4.7 Drive motor. A 28 volt (V) direct current (dc), reversible, electric drive motor shall be provided. The drive motor shall have a torque of not less than 40.6 pound-inches at 4270 revolution per minute (rpm) when operating at rated voltage. The drive motor, when installed on the actuator, shall be water resistant. The drive motor shall meet performance requirements over the full range of BSLA operating temperatures (see 4.7.3.3).

3.4.7.1 Amperage. During intermittent duty operation (20 seconds), the drive motor shall not exceed 225 amperes current draw under a 52,000 pound (lb) tensile force, under all BSLA environmental conditions (see 4.7.3.3.1).

3.4.7.2 Thermal overload protection. A self-resetting thermal-overload switch or other protective device shall be provided to prevent damage to the drive motor (see 4.7.3.3.2).

3.4.8 Manual override. Means for manual extension and retraction of the BSLA shall be provided in the event of drive motor failure (see 4.7.3.4).

3.4.9 Ventilation. The BSLA gear case shall be vented to atmosphere by means of a 0.25 to 1 pound per square inch (psi) relief valve (see 4.7.2 and 4.7.3.5).

3.4.10 Seals. Ice scraper and rubber wiper seals shall be provided to prevent the entry of water and other contaminants into the ball nut mechanism (see 4.7.2 and 4.7.3.6).

3.4.11 Oil level indicator. A sight gauge oil level indicator shall be provided to indicate the proper oil level in the BSLA gear box when the BSLA is in a stowed position, positioned 20 degrees from vertical (see 4.7.2 and 4.7.3.7).

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3.4.12 Steam and water jet cleaning. The BSLA shall withstand high-pressure steam cleaning using cleaner conforming to P-C-437 and water jet cleaning. There shall be no deterioration of or water leakage through seals or gaskets or any other defects. Paint removal shall not be a basis for rejection under this requirement. Steam and water jet cleaning are defined as follows: The jet is applied perpendicular to the surface to be cleaned not less than 1 foot from the surface for steam cleaning, and not less than 5 feet from the surface for water jet cleaning, and cleaned at a rate of 1 square foot per minute. The jet pressure shall not be less than 100 psi and not more than 110 psi (see 4.7.3.8).

3.5 Performance.

3.5.1 Dynamic load. The actuator shall be capable of exerting a force of 52,000 lb at any point along the full range of extension and retraction for a period of not less than 5 seconds (see 4.7.4.1).

3.5.2 Static load. The static tensile load capability of the BSLA shall be 105,600 lb for not less than 1 second duration (see 4.7.4.2).

3.5.3 Reverse drive. The BSLA shall not reverse drive under a 65,000 lb tensile load (see 4.7.4.3).

3.5.4 Slip clutch. The slip clutch setting shall be such that a new BSLA shall exert not less than 65,000 lb dynamic tensile load (see 4.7.4.4).

3.6 Manpower and personnel integration (MANPRINT).

3.6.1 Human engineering factors. Manual override shall be operable by 5th through 95th percentile males and females as specified in MIL-STD-1472, paragraph 5.4.4.2 (see 4.7.5.1).

3.6.2 Safety. The BSLA design shall not have any condition which may create a safety hazard to operating or maintenance personnel (see 4.7.5.2).

3.7 Environmental.

3.7.1 Environmental conditions. The BSLA, when properly serviced, shall operate under any extreme condition of inclement weather. The BSLA shall operate in ambient air temperatures from minus 50 degrees Fahrenheit (°F) to plus 120°F. When in storage, the BSLA shall withstand temperatures from minus 50°F to plus 160°F without deterioration that causes failure to any BSLA component (see 4.7.6.1).

3.7.2 Dust. The BSLA shall be subjected to one dust exposure cycle in accordance with MIL-STD-810, method 510.1. After dust exposure, the BSLA shall not be cleaned until conformance to durability requirements has been determined (see 3.8 and 4.7.6.2).

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3.8 Durability. The BSLA shall be capable of operation over a total of 500 operational durability cycles. This operational durability cycle shall also be used to verify the reliability and maintainability requirements specified (see 3.8.8, 3.8.9, and 4.7.7).

3.8.1 Operational durability cycles. Starting from the fully extended position (overall length 86 inches), the BSLA shall exert a 10,000 lb tensile force over the initial 27 inches of retraction. During the next 8 inches of retraction, the load shall be increased linearly from 10,000 to 52,000 lb. For the remaining 2 inches of retraction, the load shall be maintained at 52,000 lb. Starting with the fully retracted position (overall length 49 inches), the BSLA shall exert a 10,000 lb compressive force over the full range of extension. Auxiliary cooling of BSLA is permitted during durability test (see 4.7.2).

3.8.1.1 Cycling rate. A cycle shall consist of one full extension and full retraction of the BSLA which shall be performed in the following time frames:

Extension	1.5 minutes.
Hold	6 minutes.
Retraction	1.5 minutes.
Hold	21 minutes maximum.
Complete Cycle	30 minutes maximum.

3.8.2 Hold position. At the completion of each retraction portion of the cycle defined in 3.8.1.1, the 52,000 lb load shall be held for 10 seconds to demonstrate that the BSLA is not back driving.

3.8.3 Temperatures. The operational durability cycles shall be performed at the following ambient temperatures:

Cycle	Temperature
1 - 400	65 to 80°F
401 - 450	-50°F
451 - 500	120°F

3.8.4 Lubricant leakage.

3.8.4.1 Leakage during operation. Lubricant leakage past the external ball screw nut seal and the manual override drive seal shall not exceed 3 drops per hour. There shall be zero leakage into the drive motor. The leakage requirements shall be for all orientations of the BSLA that shall be encountered during vehicle operation (see 4.7.7.1).

3.8.4.2 Leakage during storage. There shall be zero lubricant leakage from the BSLA during horizontal storage of the BSLA with the breather vent positioned between 10 and 2 o'clock (see 4.7.7.2).

3.8.5 Scoring. There shall be no evidence of scoring of the ball screw after the BSLA has completed the operational durability cycles (see 4.7.7.3).

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3.8.6 Slip torque. The slip clutch torque shall not be degraded below a 65,000 lb output after completion of the operational durability cycles (see 4.7.7.4).

3.8.7 Water exclusion. When tested as specified in 4.7.7.5, the accumulation of water in the ballscrew tube shall be not more than 2 ounces.

3.8.7.1 Rod end sealant. A sealant conforming to MIL-A-46106 shall be applied to the rod end threads, keyway, and face of screw prior to assembly into the screw to preclude water entry in the threaded insert area of the screw (see 4.7.1, 4.7.2, and 4.7.7.5.1).

3.8.8 Reliability. The reliability of the BSLA shall have a combined mean cycle between failure (MCBF) of 2000 cycles based on an evaluation of the reliability failures that occurred when operating the BSLA under the operational durability cycles (see 4.7.7.6).

3.8.9 Maintainability. The BSLA shall have a mean time to repair (MTTR) per task of not greater than 1.0 hour (clock time) or 2.0 manhours based on an evaluation of the maintenance tasks performed under the operational durability cycles (see 4.7.7.7).

3.9 Surface finishes.

3.9.1 Chromium plating. Ball screw threads shall be chromium plated in accordance with QQ-C-320, class 2 (see 4.7.2).

3.9.2 Passivating. Stainless steel surfaces shall be passivated in accordance with QQ-P-35 (see 4.7.2).

3.9.3 Cleaning and treatment. Nonstainless surfaces shall be cleaned and treated in accordance with TT-C-490, type I or type III (see 4.7.2).

3.9.4 Painting (see 4.7.2 and 4.7.8.1).

3.9.4.1 Priming. The BSLA shall be primed per MIL-P-52192 or MIL-P-53030, except for areas noted on Drawing 11686299. The dry film thickness of the undercoat shall be 2.0 to 3.0 mils.

3.9.4.2 Topcoat. The BSLA shall be painted with chemical agent resistant coating conforming to MIL-C-46168 or MIL-C-53039 except for areas noted on Drawing 11686299. The dry film thickness of the topcoat shall be 2.0 to 3.0 mils. Color used shall be green, chip number 383.

3.10 Identification marking. All parts requiring identification shall be identified in accordance with MIL-STD-130 (see 4.7.2).

3.11 Servicing and adjusting. Prior to acceptance of the BSLA by the Government, the contractor shall adjust and service each BSLA for immediate operational use, including the following: The electrical system shall be checked for proper operation and the gear box for proper amount and grade of lubricant (see 4.7.2).

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3.12 Workmanship. Workmanship shall be of such quality as to assure that the BSLA and its components are free from defects that compromise, limit, or reduce the capability of the BSLA in the performance of its intended use (see 4.7.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10% of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 73 $\pm$ 18°F.
b. Barometric pressure 28.5^{+2}_{-3} inches mercury (Hg).
c. Relative humidity 50 $\pm$ 30%.

4.3 Classification of inspections:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   (1) Quality conformance examination (see 4.5.2).
   (2) Quality conformance tests (see 4.5.3).
c. Control tests (see 4.6).

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4.4 First article inspection. First article inspection shall be performed on preproduction or initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply BSLAs that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 In-process inspection. During fabrication of the first production BSLA, an in-process inspection will be conducted by the Government to evaluate conformance of materials and workmanship to requirements of specified documents. This inspection will be made at the contractor's or subcontractor's facility prior to the application of primer and paint. Processing and welding procedures, quality control system, and inspection records, as applicable, will be reviewed and evaluated during this inspection.

4.4.2 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of one BSLA. Preproduction inspection shall consist of inspection as specified in table I and at a location specified in the contract (see 6.2).

4.4.3 Initial production inspection. Unless otherwise specified (see 6.2), initial production inspection shall be performed. The Government shall randomly select four BSLAs for initial production inspection (see 6.2). Samples selected for initial production inspection shall be inspected as specified in table I and at a location approved by the Government.

TABLE I. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance		Control
				Examination	Tests	
Materials, design and construction	3.3 and 3.4 thru 3.4.4	4.7.1	X			
Defects	3.4.1, 3.4.3, 3.4.4, 3.4.9 thru 3.4.11, 3.8.1, and 3.9 thru 3.12	4.7.2	X	X		
Electromagnetic interference	3.4.5	4.7.3.1	X			
Ball and thread	3.4.6	4.7.3.2	X		X	X
Drive motor	3.4.7	4.7.3.3	X			X
Amperage	3.4.7.1	4.7.3.3.1	X			X
Thermal overload switch	3.4.7.2	4.7.3.3.2	X			X

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TABLE I. Classification of inspections - Continued.

Title	Requirement	Inspection	First article	Quality conformance		Control
				Examination	Tests	
Manual override	3.4.8	4.7.3.4	X			
Ventilation	3.4.9	4.7.3.5	X		X	X
Seals	3.4.10	4.7.3.6	X		X	X
Oil level indicator	3.4.11	4.7.3.7	X		X	X
Steam and water jet cleaning	3.4.12	4.7.3.8	X		X	X
Dynamic load	3.5.1	4.7.4.1	X		X	X
Static load	3.5.2	4.7.4.2	X			
Reverse drive	3.5.3	4.7.4.3	X			
Slip clutch	3.5.4	4.7.4.4	X		X	X
Human engineering factors	3.6.1	4.7.5.1	X			
Safety	3.6.2	4.7.5.2	X		X	X
Environmental conditions	3.7.1	4.7.6.1	X			
Dust	3.7.2	4.7.6.2	X			X
Durability	3.8 thru 3.8.3	4.7.7	X			X
Leakage during operation	3.8.4.1	4.7.7.1	X			
Leakage during storage	3.8.4.2	4.7.7.2	X			
Scoring	3.8.5	4.7.7.3	X			
Slip torque	3.8.6	4.7.7.4	X			X
Water exclusion	3.8.7	4.7.7.5	X		X	X
Rod End sealant	3.8.7.1	4.7.7.5.1	X			X
Reliability	3.8.8	4.7.7.6	X 1/			
Maintainability	3.8.9	4.7.7.7	X 1/			
Painting	3.9.4	4.7.8.1	X	X	X	X

1/ Not required for preproduction inspection.

4.4.4 First article inspection failure. Deficiencies found during, or as a result of, first article inspection shall be cause for rejection of the first article sample until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, first article inspection shall be evidence that all items already produced prior to completion of the first article inspection are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government will not accept products until first article inspection is completed to the satisfaction of the Government.

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4.5 Quality conformance inspections.

4.5.1 One hundred percent (100%) final inspection. Each BSLA produced shall be subjected to a complete final inspection by the contractor as depicted in table I and described in 4.5.2 and 4.5.3. Quality conformance inspection shall be conducted at a site selected by the contractor.

4.5.2 Quality conformance examination. Each BSLA shall be examined for the defects listed in table II to verify conformance of produced BSLAs to the requirements of this specification.

4.5.3 Quality conformance testing. Each BSLA shall be subjected to quality conformance in accordance with table I. Prior to test, BSLAs shall be completely assembled and serviced.

4.5.4 Examination or test failure. If any BSLA fails to pass any examination or test (see 4.5.1 thru 4.5.3) specified herein, acceptance shall be withheld until corrective action has been taken by the contractor that is acceptable to the Government.

4.5.5 Sampling. Random sampling shall not be allowed for inspections and tests referenced in this specification.

4.5.6 Recurring deficiencies.

4.5.6.1 Recurring major deficiencies. A major deficiency, table II, is recurring when the same defect occurs more than once in the same lot, or when the defect occurs in two successive lots. Recurring major deficiencies shall be cause for the entire lot(s) to be inspected for the recurring deficiencies. The deficiencies shall be corrected by the contractor when found.

4.5.6.2 Recurring minor deficiencies. A minor deficiency, table II, is recurring if it occurs more than twice in the same lot, or when the defect occurs in four successive lots. Recurring minor deficiencies shall be cause for the entire lot(s) to be inspected for the recurring deficiencies and correcting shall be accomplished prior to acceptance.

4.5.6.3 Unclassified defects. All defects that have no effect on function, safety, interchangeability or life, but that are considered departures from good workmanship shall be noted in writing. Workmanship deficiencies falling within this category and recurring in five BSLAs within a 30-day period, shall be added to the minor defects classification with no increase in AQL.

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TABLE II. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major		
101	Dimensions - out of tolerance affecting interchangeability (see 3.4.1).	SIE 1/
102	Lubricants - improper amount (see 3.4.3, 3.4.4, and 3.11).	Visual
103	Relief valve - improper valve, sticking (see 3.4.9).	Visual and functional
104	Seals - improper application, not preventing water entry (see 3.4.10).	Visual
105	Maximum retracted length of 49 inches - failure to retract completely, sticking (see 3.8.1).	Visual
106	Maximum extended length of 86 inches - failure to extend completely, sticking (see 3.8.1).	Visual
107	Sealant - improper application, not preventing water entry (see 3.8.7.1).	Visual
108	Paint - improper paint, improper application (see 3.9.4).	Visual
109	Workmanship - faulty affecting performance (see 3.12).	Visual
Minor		
201	Electrical system - inoperative, improper connections, bent or broken pins (see 3.4.1 and 3.11).	Visual
202	Lubricants - improper grade and application (see 3.4.3.1 and 3.4.3.2 and 3.11).	Visual
203	Oil level indicator - damaged (see 3.4.11).	
204	Surface finish - improper application, coverage or color (see 3.9).	Visual
205	Identification marking - missing, not legible or improper location (see 3.10).	Visual
206	Workmanship - improper appearance (see 3.12).	Visual

1/ SIE = Standard Inspection Equipment.

4.6 Control tests. Unless otherwise specified (see 6.2), control tests shall be conducted on 1 from every 50 BSLAs consecutively produced or 1 test in a 4-month period, whichever comes first. Sample shall be subjected to the control tests specified in table I except that the durability test shall consist of 50 cycles (see 3.8).

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4.6.1 Failure. Failure of the sample to pass any of the control tests specified in table I shall be cause for the Government to refuse additional testing until the faults revealed have been corrected.

4.7 Methods of inspection.

4.7.1 Materials and construction. Conformance to 3.3 and 3.4 through 3.4.4 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.7.2 Defects. Conformance to 3.4.1, 3.4.3, 3.4.4, 3.4.9 thru 3.4.11, 3.8.1, and 3.9 thru 3.12 shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with SIE.

4.7.3 Design, construction, and configuration.

4.7.3.1 Electromagnetic interference. To determine conformance to 3.4.5, the BSLA shall be tested in accordance with MIL-STD-462.

4.7.3.2 Ball and thread. To determine conformance to 3.4.6, each ball screw shall be checked for the accuracy of thread diameter and the lead. The ball shall have the specified nominal diameter. The backlash shall not exceed the specified limit.

4.7.3.3 Drive motor. To determine conformance to 3.4.7, each drive motor shall be operated for not less than 20 seconds in forward and 20 seconds in reverse direction.

4.7.3.3.1 Amperage. To determine conformance to 3.4.7.1, the current draw shall be measured under a 52,000 lb load at ambient room temperature.

4.7.3.3.2 Thermal overload protection. To determine conformance to 3.4.7.2, the thermal overload protection device shall be checked for malfunctions. A test shall be performed on the drive motor which shall cause the thermal device to open in accordance with its design. Failure of the switch or device to trip before drive motor damage occurs shall be cause for rejection.

4.7.3.4 Manual override. To determine conformance to 3.4.8, the BSLA shall be extended and retracted one time and checked for smooth operation during operation.

4.7.3.5 Ventilation. To determine conformance to 3.4.9, the gear case vent valve shall be removed and the relief pressure level measured.

4.7.3.6 Seals. To determine conformance to 3.4.10, the functional operation of the ice scraper and rubber wiper seals shall be verified.

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4.7.3.7 Oil level indicator. To determine conformance to 3.4.11, a check shall be made to determine that an oil level indicator was provided and reflects the proper operating oil level.

4.7.3.8 Steam and water jet cleaning. To determine conformance to 3.4.12, the BSLA shall be cleaned as specified. There shall be no deterioraton of performance due to cleaning.

4.7.4 Performance.

4.7.4.1 Dynamic load. To determine conformance to 3.5.1, the BSLA shall be cycled once at a 52,000 lb load.

4.7.4.2 Static load. To determine conformance to 3.5.2, the BSLA shall be subjected to a 105,600 lb tensile load. Structural integrity shall not be degraded as determined by a visual inspection of the complete assembly upon completion of this test.

4.7.4.3 Reverse drive. To determine conformance to 3.5.3, the BSLA shall be subjected to a 65,000 lb tensile load. The BSLA shall not reverse drive at the specified load.

4.7.4.4 Slip clutch. To determine conformance to 3.5.4, the BSLA shall be subjected to a 65,000 lb dynamic tensile load for 2 seconds. The slip clutch shall not slip at the specified load.

4.7.5 MANPRINT evaulations.

4.7.5.1 Human engineering factors. To determine conformance to 3.6.1, the torque required to operate the manual override shall be measured and checked for conformance to capabilities of 5th through 95th percentile males and females as specified in MIL-STD-1472.

4.7.5.2 Safety. To determine conformance to 3.6.2, the BSLA shall be inspected for safety hazards.

4.7.6 Environmental.

4.7.6.1 Environmental conditions. To determine conformance to 3.7.1, the BSLA shall be operated and stored through the temperature range as specified.

4.7.6.2 Dust. To determine conformance to 3.7.2, the BSLA shall be subjected to dust contamination prior to the start of the durability test.

4.7.7 Durability. To determine conformance to 3.8, the BSLA shall be subjected to an operational durability test of 500 cycles. The loads, cycling rates, holding load, and temperatures shall be as specified in 3.8.1 through 3.8.3.

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4.7.7.1 Leakage during operation. To determine conformance to 3.8.4.1, the lubricant leakage shall be observed during the durability test. Leakage shall not exceed 3 drops of gear case oil.

4.7.7.2 Leakage during storage. To determine conformance to 3.8.4.2, the BSLA shall be stored for 24 hours in the horizontal position with the breather vent located between the 10 and 2 o'clock position. No leakage of lubricant shall be observed.

4.7.7.3 Scoring. To determine conformance to 3.8.5 and after the completion of the test specified in 4.7.7, the entire length of the ball screw shall be checked for any evidence of scoring using 10X magnification inspection.

4.7.7.4 Slip torque. To determine conformance to 3.8.6, after the completion of the test specified in 4.7.7, the BSLA shall be operated at a 65,000 lb tensile load for 2 seconds. The slip clutch shall not slip.

4.7.7.5 Water exclusion. To determine conformance to 3.8.7, a water seal test shall be performed after completion of the operational durability cycle test. The test shall be performed with the BSLA extended 20 inches from the rod end and the ball screw actuator in a vertical position pointing up. The BSLA shall be subjected to a constant flow of water running down the screw at a rate of 1 gallon per hour for 1 hour. Accumulation of water in the ballscrew tube in excess of 2 ounces is unacceptable. Water accumulation shall be determined by removing the breather vent and any grease blocking the hole and then inverting the actuator and measuring the amount of water that runs out of the breather vent.

4.7.7.5.1 Rod end sealant. To determine conformance to 3.8.7.1, the rod end shall be checked to verify that sealant is applied as specified.

4.7.7.6 Reliability. To determine conformance to 3.8.8, a reliability verification of the 2000 MCBF of the BSLA shall be conducted by testing 4 BSLAs for 500 operational durability cycles each in accordance with 3.7.2 and 3.8 through 3.8.3. The 2000 MCBF requirement shall be considered met if there are no failures after the 4 BSLAs have completed a cumulative total of 2000 cycles.

4.7.7.7 Maintainability. To determine conformance to 3.8.9, the MTTR per task of 1.0 hour (clock time) or 2.0 manhours shall be verified in conjunction with the reliability test (see 4.7.7.6).

4.7.8 Surface finishes.

4.7.8.1 Painting. To determine conformance to 3.9.4, perform the film thickness and scribe tape tests specified in MIL-STD-193.

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5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The BSLA covered by this specification is used for deployment and retraction of the outriggers of M860A1, PATRIOT Radar and Launcher System.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If first article inspection is not required (see 3.1).
c. If responsibility for inspection shall be other than as specified (see 4.1).
d. If responsibility of inspection equipment shall be other than as specified (see 4.1.2).
e. If inspection conditions shall be other than as specified (see 4.2).
f. If preproduction inspection is required (see 4.4.2).
g. Location for conducting preproduction inspection (see 4.4.2).
h. If initial production inspection is not required (see 4.4.3).
i If control tests are not required (see 4.6).
j. Selection of applicable level and packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.3.2).

6.3.2 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.4 Subject term (key word) listing.

Actuator, linear, ball screw
Linear actuators, ball screw

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6.5 Supersession data. This military specification supersedes ATPD 2086, dated 6 June 1986.

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project 5945-A668)

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